Agenda

Sunday, October 13th

4-6:30PM
Arrival and Check-in

7-8PM
Plenary Lecture, David Tobin, PhD, Duke University
" From Zebrafish to Humans: Understanding How Pathogens Reprogram Immunity"

8PM
Reception with heavy Hors d'oeuvres
Cash Bar

Monday, October 14th

8:00-9:00 AM
Breakfast

9:00-10:30 AM
Session 1: Chemotaxis
Chair: Cynthia Damer, Central Michigan University

9:00-9:30: Jeffery Hadwiger, Oklahoma State University
Function and Regulation of the Atypical MAPK Erk2

9:30-10:00: Jonathan Moore, Pomona College
A Rich Long-Running Inquiry-Based Introductory *Dictyostelium* Chemotaxis Teaching Lab

10:00-10:30 Xuehau Xu, NIAID-NIH
RasGAP Control G protein coupled receptor (GPCR)-mediated Gradient Sensing Dynamics in *Dictyostelium*

10:30-11:00 AM
Coffee Break

11:00-12:30 PM
Session 2: Development
Chair: Michelle Kovarik, Trinity College

11:00-11:25: Julie Hesnard, Institut Lumière Matière
Transcriptional hypoxic response and directed oxygen migration in amoebae

11:30-11:55: Chris West, University of Georgia
Proteomic studies in search of mechanisms of oxygen regulation of aerotaxis and culmination in *Dictyostelium*

12:00-12:25: Fu-Sheng Chang, NIDDK-NIH
A Cell-Density Sensing Factor with Secreted, Non-Autonomous and Cell Autonomous Functions that Regulate *Dictyostelium* Growth and Development

12:30-2:00 PM
Lunch

2:00-3:00 PM
Session 3: Host Microbe Interactions
Chair: Xuehua Xu, NIAID-NIH

2:00-2:25: Joel Ching, University of Jena
Tracking the dynamics of interacting microbial populations

2:30-2:55: Tera Levin, University of Pittsburgh
An ideal set of Dictyostelium sister-species genomes to assess conservation and selection

3:00-4:00 PM
Break

4:00-5:30 PM
Session 4: Cellular Signaling
Chair: Charest Pascale, University of Arizona

4:00-4:25: Jonathan Chubb, University College London
Group allocation during aggregation and beyond

4:30-4:55: Yusuke Morimoto, Kyushu Institute of Technology
Fluorescence microscopy measurement of ion signaling in *Dictyostelium discoideum* cells

5:00-5:25: Pundrik Jaiswal, NIDDK-NIH
Identification of Pterin Receptors and their Signaling Pathways in Dictyostelium

5:30-6:30 PM
Break

6:30-8:00 PM

Dinner

8:00-9:30 PM

Poster Session 1 (Even Numbers Present)

Tuesday, October 15th

8:00-9:00 AM
Breakfast

9:00-10:30 AM
Session 5: Chemotaxis II
Chair: Robert Huber, Trent University

9:00-9:30: Pascale Charest, University of Arizona
Studying the interaction of cAR1 with G protein subunits by BRET

9:30-10:00: Christopher Thompson, University College London
Collective oscillatory signaling acts as a developmental timer initiated by weak coupling of a noisy pulsatile signal

10:00-10:30 Wouter-Jan Rappel, University of California San Diego
Eukaryotic chemotaxis under periodic stimulation shows temporal gradient dependence

10:30-11:00 AM
Break

11:00-12:30 PM
Session 6: Computational Resources
Chair: Tera Levin, University of Pittsburgh

11:00-11:25: Petra Fey, Northwestern University
An update on the Dictyostelium Community Resource

11:30-12:30: Open Discussion on the Future of Computational Resources
Moderators: Tera Levin, Matt Scaglione, Jonathan Chubb

12:30-2:00 PM
Lunch

2:00-3:30 PM

Session 7: Host Microbe II:
Chair: Kari Naylor, University of Central Arkansas

2:00-2:25: Emily Cruz-Lorenzo, University of Pittsburgh
Uncovering antibacterial functions of TirA, a TIR-domain containing protein in Dictyostelium discoideum

2:30-2:55: Mische Holland, University of Pittsburgh
Novel families of Chlamydiae bacteria form stable facultative endosymbiosis with wild Dictyostelium species

3:00-4:00 PM

Break

4:00-5:30 PM

Session 8: Dictyostelium to Model Disease
Chair: TBD

4:00-4:25: Robert Huber, Trent University
The pre-mRNA splicing kinase Prp4K modulates autophagy in *Dictyostelium*

4:30-4:55 Felicia Williams, Duke University
Acute Stress and Multicellular Development Alter the Solubility of the *Dictyostelium* Sup35 Ortholog ERF3

5:00-5:25: William Kim, Trent University
Using *Dictyostelium* to study the molecular impacts of disease-causing mutations in *CLN5*

5:30-6:30 PM

Break

6:30-8:00 PM

Dinner

8:00-9:30 PM

Poster Session II (Odd Numbers Present)

Wednesday, October 16th

8:00-9:00 AM
Breakfast

9:00-10:30 AM
Session 9: Development II
Chair: TBD

9:00-9:25: Gad Shaulsky, Baylor College of Medicine
Allorecognition, altruism and cheating in the *tgrB1-tgrC1-rapgapB* greenbeard pathway

9:30-10:00: Satoshi Kuwana, The University of Tokyo
TgrN1 mediates cell intercalation in D. discoideum

10:00-11:00 AM
Break

11:00-12:30 PM
Session 10: Cytoskeleton
Chair: TBD

11:00-11:20: Allyson Sygro, HHMI-Janelia Campus
“TBD”

11:30-11:55: Yu Deng, Johns Hopkins University
PIP5K-Ras bistability initiates plasma membrane symmetry breaking to regulate cell polarity and migration

12:00-12:25 Cody Morrison, Central Michigan University
Copine A (CpnA) promotes the dimerization and activation of PatA, a Ca^{2+}-ATPase, in *Dictyostelium*

12:30-1:00 PM
Pick up to go lunch

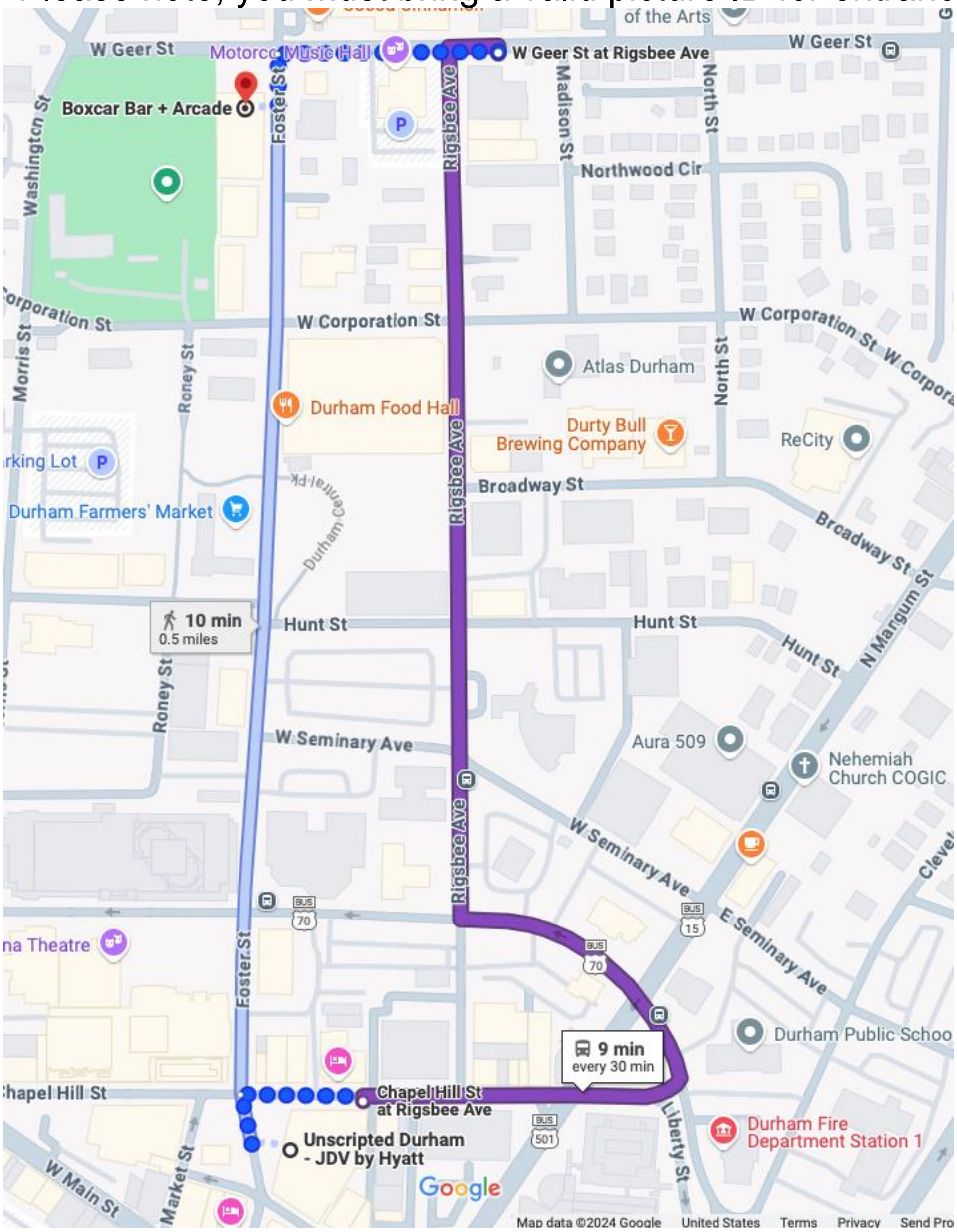
1:00 PM
Load bus for travel to either Eno River Trail or Duke Gardens

Agenda

4:30 PM Return

6:00-10:00 PM

Dinner and Games at BoxCar Arcade and Bar
Meet at 621 Foster St, Durham NC, 27701
*Please note, you must bring a valid picture ID for entrance



Thursday, October 17th

Depart

Abstracts
Oral Presentations

Function and Regulation of the Atypical MAPK Erk2

Ramee Aranda, Md. Ikram Rafid, Saher Fatima and Jeffrey Hadwiger*
Department of Microbiology and Molecular Genetics, Oklahoma State University

Abstract

Understanding the regulation and function of atypical mitogen-activated protein kinases (MAPKs), such as homologs of mammalian MAPK15, have been a challenge in most eukaryotes due to the lack of identified external activating signals. In *Dictyostelium*, the atypical MAPK Erk2 is activated and required for chemotactic responses to folate and cAMP, including chemotactic movement and transcription factor regulation. Activated Erk2 phosphorylates a key developmental transcription factor, GtaC, leading to the translocation of GtaC from the nucleus to the cytoplasm. Both foraging and developmental signals lead to Erk2 regulation of GtaC translocation suggesting that other transcription factors might provide cell fate specific gene regulation. Atypical MAPKs are not activated by conventional upstream MAPK kinases as confirmed in *Dictyostelium* by the activation of Erk2 in cells lacking the only known MAPK kinase, MekA. Using a kinase-dead version of Erk2 we find that this functionally inactive protein is phosphorylated in response to chemoattractants suggesting an unknown protein kinase acts upstream to regulated Erk2. We are exploring the possible roles of other protein kinases in the activation of Erk2. Using a chimeric MAPK analysis we found a conserved motif in the carboxyl terminus that is important for Erk2 function. These findings suggest atypical MAPKs have important roles in chemotaxis and cell fate.

*Presenter

A Rich Long-Running Inquiry-Based Introductory *Dictyostelium* Chemotaxis Teaching Lab

Jonathan E. Moore[a]
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Over eleven years, more than 1100 mostly first-year Pomona College students have performed an inquiry-based teaching lab module on *Dictyostelium* chemotaxis using modified versions of the under-agarose assay of Laevsky and Knecht (2001). After a dry run of the assay and subsequent analysis of their time-lapse video, each pair of students searches the scientific literature for an idea as to how to change the assay to investigate an aspect of chemotaxis. Subsequently, they perform their proposed experiment, analyze it, and present their findings to their peers and instructors. Over the five to seven lab sessions, students develop their microscopy skills, pipetting, dilutions calculations, notebook keeping, and many analytical and statistical skills. The lab also helps to reinforce classroom concepts in cell motility, the cytoskeleton, and signal transduction.

RasGAP Control G protein coupled receptor (GPCR)-mediated Gradient Sensing Dynamics in *Dictyostelium*

Xuehua Xu
Chemotaxis Signaling Section, Laboratory of Immunogenetics, National Institute of Allergy and Infectious Diseases, National Institutes of Health, 5625 Fishers Lane, 4N03, Rockville, MD 20852, USA.

ABSTRACT
Dictyostelium is the best-known system for G protein coupled receptor (GPCR)-mediated chemotaxis. Ras plays a central role in multiple signaling transduction pathways by which *Dictyostelium* cell controls the directional cell migration. Ras is a molecular switch and its activity is regulated by GEFs and GASPs. In recent years, we identified two Ras GAPs (C2GAP1 and C2GAP2) and revealed their essential roles in the GPCR-mediated Ras adaptation, polarization, and chemotaxis in *Dictyostelium*. However, molecular mechanism of the RasGAP in regulating the dynamics of gradient sensing during chemotaxis largely remain elusive. In the present study, we investigated the role of C2GAP1 in mediating cAMP concentration-dependent dynamics of gradient sensing.

Transcriptional hypoxic response and directed oxygen migration in amoebae

Julie Hesnard[1], Satomi Hirose[2*], Nasser Ghazi[1], Mitchell Rich[3], Hanke van der Wel[3], Olivier Cochet-Escartin[1], Jean-Paul Rieu[1], Christopher M. West[3], Christophe Anajrd[1]
[1] Institut Lumière Matière, University of Lyon, Université Claude Bernard Lyon 1, CNRS, Villeurbanne, France
[2] Institute of Fluid Science, Tohoku University, Sendai, Japan
[3] Department of Biochemistry and Molecular Biology, University of Georgia, Athens, GA, United States
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In many organisms, oxygen is one of the main growth limiting factors. As the level of available oxygen can fall rapidly, cells have mechanisms to adapt to this hypoxic situation. A first strategy employed by some cells is to move towards an optimal oxygen concentration, a phenomenon known as aerotaxis. This phenomenon was first observed and studied in bacteria, but recent studies showed that it is also involved in numerous physiological processes, such as morphogenesis and cancer cells spreading. However, the mechanism involved in aerotaxis remain unknown. A first part of our work focuses on the molecular characterisation of aerotaxis. We developed an oxygen migration assay that showed that *Dictyostelium Discoideum* processes aerotactic capacities (1). Using this assay with null mutants, we determined that several genes of the oxygen recognition pathway, are not involved in aerotaxis. We also reported that mitochondrial activity, oxidative and nitrosative stresses are not required for aerotaxis (2).
Another strategy used to adapt hypoxia is genic adaptation. The only well-described adaptation system is the metazoans PHD (prolyl hydroxylase)/HIF-1 (Hypoxia-Inducible factor) pathway, which regulates the transcription of around a hundred genes. However, the transcriptional changes at the level of unicellular cells or in other eukaryotic organisms are poorly understood. The second part of our work focuses on the characterisation of the transcriptional hypoxic regulation. For that, we realised a RNA-seq time course with cells placed at 1% oxygen and then reoxygenated. This experiment was coupled with proteomic analyses performed by the Pr. West team. By analysing the transcriptomic data, we reveal a huge induction of genes related to ribosome biogenesis after 1 hour in hypoxia. After 24 hours in hypoxia, we determined that genes involved in cellular compound degradation (endocytosis, autophagy) and in cytoskeleton organisation were overexpressed. RT-qPCR analyses were performed to complement the RNA-seq. We showed that the hypoxic response can occur within minutes for some tested genes (<10 min). We also demonstrated that the response is dependent on the oxygen concentration, with an induction that stared at 15% O_2 and that increased as the O_2 concentration decreases. Finally, we showed that the induction is mediated by multiple mechanisms.

References:

1. Cochet-Escartin O et al. Hypoxia triggers collective aerotactic migration in *Dictyostelium discoideum*, eLife, 2021.
2. Hirose S, Hesnard J et al. The aerotaxis of *Dictyostelium discoideum* is independent of mitochondria, nitric oxide and oxidative stress. Front Cell Dev Biol, 2023.

Key words: hypoxia, transcriptional response, aerotaxis, *Dictyostelium discoideum*, RNA-seq

Proteomic studies in search of mechanisms of oxygen regulation of aerotaxis and culmination in *Dictyostelium*

Hanke van der Wel[1], Elisabet Gas-Pascual[1], Julie Hesnard[2], Christophe Anjard[2], Christopher M. West[1]

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Like most aerobic organisms, *Dictyostelium* responds to temporal and physical gradients of O_2 concentration for positional information and metabolic adaptation. For example, a threshold of around 10-12% O_2 is required for commitment of a slug to culminate into a fruiting body. Our past studies identified the Skp1 prolyl hydroxylase PhyA as a molecular component for sensing levels of O_2 sufficient to permit culmination at an air-water interface and sporulation in a submerged assay. Our finding that absence of PhyA can be overridden by high O_2 levels indicates the occurrence of additional molecular sensing mechanisms. In search of other candidates, we performed a transcriptomic time course analysis of development from 10-18 h at 21% O_2 (atmospheric level) and 5% O_2 (mimicking the soil environment). Minimal differences up to 10 h were followed by over 500 differentially expressed genes. However, interpretation is confounded by the dynamic nature of the developmental transcriptome. Therefore as proxy, we turned to vegetative stage cells and their proteomic response to a switch from 21% to 1% O_2. Of around 2000 proteins that could be confidently quantified, over 50 were increased and 14 were decreased by >30% after 5 h. Many were anticipated by changes at 1 h and maintained at 24 h. Most of the changes were coordinate with findings from a simultaneous transcriptome analysis by Julie Hesnard and Christophe Anjard in Lyon. Many of the proteins that increased were O_2-dependent enzymes suggesting a homeostatic response. Several of the proteins have regulatory potential and are under further investigation. A detailed analysis of the changes will be correlated with the vegetative and developmental transcriptomes and summarized at the meeting.

A Cell-Density Sensing Factor with Secreted, Non-Autonomous and Cell Autonomous Functions that Regulate *Dictyostelium* Growth and Development

Fu-Sheng Chang[1], Pundrik Jaiswal[1], Geoffrey Holm[1,2] and Alan R Kimmel[1]
1. National Institute of Diabetes and Digestive and Kidney Diseases, National Institutes of Health, Bethesda, MD 20892
2. Department of Biology, Colgate University, Hamilton, NY 13346

Developmental aggregation of *Dictyostelium* does not occur at low cell densities, However, cells developed in the presence of conditioned media from high-density developed cells will aggregate at non-permissive cell densities. Using low cell density aggregation as an assay, we purified a secreted 150 kDa protein (p150) with aggregation activity. Mass Spectrometry (MS/MS) peptide sequence analysis was used to identify the gene *DPF1* (Developmental Promoting Factor 1). Cells that overexpress DPF1 aggregate at lower cell densities than do WT cells and *dpf1*-null cells, which lack the secreted aggregation activity, only aggregate at cell densities higher than that required by WT. Tag-affinity purified secreted DPF1 possesses density aggregation activity and cells that overexpress DPF1 aggregate at lower cell densities than do WT cells. In mixed studies with WT cells, cells overexpressing DPF1 rescue low density aggregation, define centers for multi-cell aggregation, and determine cell-fate choice during *Dictyostelium* cytodifferentiation.

DPF1 is initially synthesized as a larger (160 kDa) single-pass transmembrane (TM) protein, that undergoes proteolytic cleavage and ectodomain shedding to release p150 with aggregation activity. A remaining ~10 kDa TM/cytoplasmic segment of DPF1 functions independently, promoting cell-substratum adhesion and cellular growth. Secreted p150 DPF1 possesses an Epidermal Growth Factor (EGF)-like domain, and the TM region of DPF1 has strong homology with EGF Receptor (EGFR) motifs, although it lacks a kinase domain in its cytoplasmic segment.

Our research also led to the discovery of DPF2, which shares close sequence similarity with DPF1, including the EGF-like domain and the EGFR motif, and exhibits comparable processing and ectodomain shedding properties. We have purified secreted p150 from both DPF1 and DPF2, analyzed them by MS/MS, and identified their unique ectodomain sheddase proteolytic cleavage sites; ectodomain cleavage is largely dependent upon calcium and calcium-dependent proteases (calpains).

We have also shown homo- and hetero-dimerization of the transmembrane (TM/cyto) domains of both DPF1 and DPF2, but also physical interaction between the secreted 150 kDa fragments and the TM/cyto domains, suggesting ligand/receptor function. Dose-dependent assays of aggregation, using DPF1 and DPF2 alone or in mixes, show cooperativity between DPF1 and DPF2.

Finally, we have applied Alphafold modelling for 3D structural analyses of DPF1 and DPF2. The sheddase cleavage sites are located at the edge of a beta sheet, suggesting that calpain recognizes the cleavage site by its conformation. Deletion of sequences adjacent to the cleavage site induce conformational changes in the beta sheet and inhibits processing, perhaps by limiting accessibility to calpains, but also dimerization. Deletion of the EGF domains in the DPFs appears to cause conformational changes that also influence the dimerization and processing of the full-length protein. TM/cyto dimerization and processing are tightly coupled; interference of proteolytic cleavage and ectodomain shedding disrupts TM/cyto dimerization.

Tracking the dynamics of interacting microbial populations

Joel Ching1,2,3, Silvia Vareschi1,2,3, Marco Mauri1,2, Ariane Zander1,2, Tatjana Malycheva2,5, Nia Verdon1,2, Rosa Herbst2,4, Stefan Schuster2,5, Pierre Stallforth2,3,4, Rosalind J. Allen1,2,3
1 - Theoretical Microbial Ecology Group, Institute of Microbiology, Friedrich Schiller University Jena, Germany
2 - Cluster of Excellence "Balance of the Microverse", Jena, Germany
3 - Jena School for Microbial Communication, Jena, Germany
4 - Leibniz Institute for Natural Product Research and Infection Biology, Jena, Germany
5 - Matthias-Schleiden-Institut, Friedrich Schiller University Jena, Germany

Microbes exhibit diverse interactions including predation, symbiosis, and competition. These interactions impact the dynamics of microbial populations and therefore shape ecological outcomes within microbial communities. This study focuses on the interactions between *Dictyostelium discoideum* (Dicty) and bacteria. Dicty is a social amoeba that feeds on soil bacteria by phagocytosis. During starvation, it develops into multicellular aggregates and forms fruiting bodies. It has been reported that some Pseudomonads produce toxic compounds that prevent Dicty predation. This study seeks to understand predatory, foraging, and defense behaviours between Dicty and different bacteria species. We hypothesize that the dynamics of these interactions are largely dependent on the population sizes and spatial distribution of the interacting microbes.
We developed experimental setups that allow microscopic imaging of the interaction between Dicty and bacteria in liquid suspension and on agar. The images are analyzed using ImageJ/Mtrack/Labkit and MATLAB. The results obtained will ultimately be compared with mathematical models.
By combining population dynamics experiments with mathematical models, we intend to provide a better insight into how interactions mediated by natural products can affect the balance of microbial ecosystems. This understanding would help develop solutions to ecological, agricultural, clinical, or environmental problems, as well as the advancement of natural product research.

Title: An ideal set of Dictyostelium sister-species genomes to assess conservation and selection

Authors: Mische Holland, May Ahmad, Tera Levin

Affiliations: University of Pittsburgh, Dept of Biological Sciences

Evolutionary genomic analyses of conservation and selection require genomes that are different enough that there is genetic variation to observe, but not so different that the genomic record has been repeatedly overwritten by mutation. In Dictyostelium, these evolutionary analyses have been complicated by the fact that many D. discoideum strains are extremely genetically similar, while the closest well-sequenced relatives such as D. purpureum are too diverged. To address this gap, we used long-read and short-read methods to sequence, de novo assemble, and annotate a suite of Dictyostelium genomes of an ideal divergence for comparative genomics analyses. We present a new reference Ax2-214 genome as well as nine additional genomes from strains that are likely sister-species to D. discoideum. We discovered chromosomal re-arrangements between closely related Dictyostelium spp. as well as gene family expansions and rapidly evolving gene sequences. We present the family of TIR-domain-containing genes as an illustrative example. Overall, we expect these genomes will be a very useful resource for any researchers seeking to analyze genetic conservation or divergence in Dictyostelium.

Group allocation during aggregation and beyond

Hugh Z Ford, Paco Chow, Ian Jones and Jonathan R Chubb

Laboratory for Molecular Cell Biology, University College London, UK

Abstract:
I will outline our approaches to determine how cell populations are partitioned into different aggregates following starvation. Using centimetre-scale fields of view, we imaged the emergence, stabilisation and pruning of signalling centres during the onset of collective cAMP signalling. The number of centres in a given area follows a reliable scaling with population size, implying homeostatic regulation of group size. We explored the rules of this homeostasis using a cAMP signalling reporter and automated detection of nascent centres (singularities). New singularities emerge within a short (20 minute) window prior to collective chemotaxis, followed by pruning of singularities by three different disintegration mechanisms. A window of competence for establishment of singularities was revealed by optogenetic a) culling or b) ectopic initiation of centres.

I will also describe our approaches to determine cell state trajectories during development. Although it is known that initial cell state (eg. cell cycle, nutrition) can bias final cell fate, it is unclear what intermediate cell states cells transit through depending on the initial cell state. Using pooled analysis of different single cell RNAseq datasets, we have defined gene expression features of candidate cell state trajectories branching from the undifferentiated cell populations. We are now testing the validity of these predicted cell state trajectories using cross scale imaging of multiple readouts of cell state.

Agenda

Fluorescence microscopy measurement of ion signaling in *Dictyostelium discoideum* cells

Yusuke V. Morimoto: yvm001@phys.kyutech.ac.jp
Yusuke V. Morimoto[1,2], Yukihisa Hayashida[1], and Masahiro Ueda[2,3]

[1] Graduate School of Computer Science and Systems Engineering, Kyushu Institute of Technology, Fukuoka, Japan.
[2] RIKEN Center for Biosystems Dynamics Research (BDR), Osaka Japan.
[3] Graduate School of Frontier Biosciences, Osaka University, Osaka, Japan.

Cell-to-cell communication is fundamental to the organization and functionality of multicellular organisms. Intercellular signals orchestrate a variety of cellular responses, including gene expression and protein function changes. *Dictyostelium discoideum* is a model organism for cell-to-cell interactions mediated by chemical signals and multicellular formation mechanisms. Upon starvation, *D. discoideum* cells exhibit cAMP gradients and chemotaxis, which facilitates the unicellular-to-multicellular transition. During this process, the calcium ion signaling synchronizes with the cAMP signaling. The fact that there is ion flow suggests that the cell membrane potential changes. On the other hand, in the multicellular formation phase, decrease in cytosolic pH caused cell differentiation into stalk cells. These suggest that ion signaling is at work at each stage of *D. discoideum*. Unlike biosynthesized signals, ions are absorbed from the environment, making them versatile and essential signaling factors for organisms. Advantageously, ion-mediated signals, such as action potentials in neurons, can be transmitted rapidly and over long distances. To investigate the role of ion signaling, we have established highly sensitive fluorescence imaging techniques for Ca^{2+}, pH, and membrane potential associated with cell dynamics in both the unicellular and multicellular stages. I will discuss the role of ion signaling as revealed by each ion imaging.

Identification of Pterin Receptors and their Signaling Pathways in Dictyostelium.

PUNDRIK JAISWAL, NETRA PAL MEENA, ALAN R KIMMEL
National Institutes of Diabetes and Digestive and Kidney Diseases, National Institutes of Health. Bethesda, Maryland 20892, USA.

Folate and related Pterins are a class of heterocyclic compounds that serve as a crucial cofactors in complex biological processes throughout the Eukarya. Folate released by bacteria are chemoattractants for growing Dictyostelium in nutrient searching, but other pterin derivates (e.g. monapertin) are suggested to be essential for the development of Dictyostelium multicellularity and cell fate decisions. We and others have described Folate Receptor (FolR)/Ga4 coupled events in the context of chemotaxis and ERK2 activation. We have now extended these studies to understand structural elements within the Pterin family that are essential for receptor binding and activation. We have also described the biochemical basis to folate de-activation by deamination and have identified activating Pterin family members that are resistant to de-amination. In addition, we have searched for additional receptors that are sensitive to stimulation by Pterin family members, using developmental times when FolR is not expressed. ERK2 is rapidly activated in growing Dictyostelium in response to exogenous folate stimulation, however, this ERK2 response is lost during progression through early development. FolR, a member of a group of sequence-related GPCRs, yet FolR is the primary receptor required for high activation of ERK2 by folate. We expressed a series of FolR-related receptors during development of WT and folR-null cells and monitored ERK2 activation by folate. Two novel Pterin-sensitive receptors (PtRs) were identified that further required Ga4 for ERK2 activated-coupling. We then compared multiple folate response readouts of WT, folR-null, folRnull/ PtR-overexpressors, PtR-overexpressors, and PtR-null cells and confirmed a complex interaction of FolR and PtRs in the Folate signal-response. These novel Pterin receptors respond to folate and monapterin with high affinity and show increased phosphorylation in response to folate. We are using Mass Spectrometry analyses to study folate-dependent phosphorylations and multi-protein interactions.

Agenda

Studying the interaction of cAR1 with G protein subunits by BRET

Helena A. Woroniecka[1], AFM Tariqul Islam[1], and Pascale G. Charest[1,2,3,4]
[1] Department of Chemistry and Biochemistry, U. Arizona, Tucson AZ
[2] Department of Molecular and Cellular Biology, U. Arizona, Tucson AZ
[3] U. Arizona Cancer Center, Tucson AZ
[4] BIO5 Institute, Tucson AZ

We recently adapted the use of Bioluminescence Resonance Energy Transfer (BRET) methods to study the real-time interaction of proteins in live *Dictyostelium* cells. We successfully used BRET to monitor G2 protein subunit dissociation and detect the activation of G2 by the cAMP chemoattractant receptor cAR1 in response to cAMP. Now, using cAR1-h*R*LucII, Galpah2GFP2, and Gbeta-GFP2, we report the detection and monitoring of the interaction between cAR1 and the individual G2 protein subunits. We observed basal BRET signals between the G2 subunits and cAR1 in resting cells and whereas cAMP stimulation increases the BRET signals detected between the receptor and both Galpha2-GFP2 and Gbeta-GFP2, it does so with distinct dynamics and effective cAMP concentrations. Together with previous studies that characterized the dynamic localization of cAR1 and the G2 protein subunits in response to cAMP stimulation in *Dictyostelium*, the detailed description of the interaction dynamics between them will help understand the early steps involved in chemoattractant gradient sensing.

Collective oscillatory signaling acts as a developmental timer initiated by weak coupling of a noisy pulsatile signal

Christopher A. Brimson1, Robert Baines1, Elisabeth Sams-Dodd1, Ioanina Stefanescu1, Bethany Evans1, Satoshi Kuwana2, Hidenori Hashimura2, Satoshi Sawai2 and Christopher R.L. Thompson1*

1 Centre for Life's Origins and Evolution, Department of Genetics, Evolution and Environment, University College London, Darwin Building, Gower Street, London, WC1E 6BT, UK
2 Graduate School of Arts and Sciences, University of Tokyo, Komaba, Meguro-ku, Tokyo, Japan

Oscillatory phenomena are ubiquitous across different biological scales, from individual cells to tissues and whole organisms. In physical systems, oscillatory behaviour can emerge from weak coupling of independent stochastic elements. Once established, oscillations can drive coordinated group behaviour and encode robust information through oscillation frequency or amplitude. We have used experimentation and modelling to understand the emergence and information coding properties of oscillatory cyclic adenosine monophosphate (cAMP) signaling in *D. discoideum*. Oscillatory cAMP signaling drives aggregation of individual amoebae and coordinated temporal shifts in gene expression. However, it is unknown how the collective behaviour emerges and attenuates in an auto-regulatory manner where temporal transcriptional regulation must decode variation in cAMP pulse count, frequency, or amplitude. To address these questions, we identified a transcription factor, Hbx5, that is essential for the initiation of collective behaviour and its activity provides a single-cell readout of the earliest sign of cAMP signaling. This reveals stochastic pulses in cAMP signaling precede collective oscillations, with Hbx5 dependent transcriptional feedback enhancing signal sensitivity necessary for the emergence of collective oscillations. We demonstrate that once collective oscillations have been established, another cAMP-regulated transcription factor, GtaC, required for subsequent developmental progression becomes activated. Modelling and experimentation suggest that temporal differences in transcription factor activation arise from differences in the threshold at which the protein kinase, ErkB, influences their activity. This leads us to propose that changes in the amplitude of cAMP oscillations during development encode temporal information, driving the orderly progression of development. These studies thus provide crucial insights into the principles driving collective behaviour and the evolution of multicellular systems.

Title: Eukaryotic chemotaxis under periodic stimulation shows temporal gradient dependence

Agenda

Authors: Richa Karmakar, Aravind Karanamd, Man-Ho Tang, and Wouter-Jan Rappel

Department of Physics, University of California San Diego

Abstract: When cells of the social amoeba Dictyostelium discoideum are starved of nutrients they start to synthesize and secrete the chemical messenger and chemoattractant cyclic Adenosine Mono Phosphate (cAMP). This signal is relayed by other cells, resulting in the establishment of periodic waves. The cells aggregate through chemotaxis towards the center of these waves.
We investigated the chemotactic response of individual cells to repeated exposure to waves of cAMP generated by a microfluidic device. When the period of the waves is short, the chemotactic ability of the cells was found to increase upon exposure to more waves, suggesting the development of a longer-term memory. This effect was not significant for longer wave periods.
We show that the experimental results are consistent with a model that includes a slowly rising and decaying component that is activated by the temporal gradient of cAMP concentration.
The observed enhancement in chemotaxis is relevant to populations in the wild: once sustained, periodic waves of the chemoattractant are established, it is beneficial to cells to improve their chemotactic ability in order to reach the aggregation center sooner.

Petra Fey, Research Associate, Center of Genetic Medicine Department (CGM)
Senior Biocurator and Stock Center Manager
Northwestern University, Chicago

Agenda

I'm a Senior Biocurator at dictyBase. I also manage the Dicty Stock Center and find new technicians if necessary. I curate papers when I have time and add GO and phenotypes in a table for the new database. I create and edit all our internal ontologies, such as phenotypes, environment and assay for example. I also took all the community annotation from the old dictyBase and entered those in Github as Markdown to make it easy to show them in a tab in the new database https://github.com/dictyBase/community-annotations/wiki. I help with the new database where I can and Bob the other curator, we add phenotype tables to be entered into the new database. GO is added in Protein2GO and shows up in QuickGO also when added by UniProt or others, but I can just show those from dictyBase, which are 18,647 annotations including ND (no data) and ISS (inferred from others) When we select only from experimental annotations the number is currently 9,473. https://www.ebi.ac.uk/QuickGO/annotations?taxonId=44689&taxonUsage=descendants&assignedBy=dictyBase&evidenceCode=ECO:0000269&evidenceCodeUsage=descendants

Title: Uncovering antibacterial functions of TirA, a TIR-domain containing protein in Dictyostelium discoideum

Agenda

Authors: Emily Cruz-Lorenzo, Edward Culbertson, Tera Levin

Affiliations: University of Pittsburgh, Dept of Biological Sciences

Innate immunity is the first line of defense for organisms across the tree of life, which are constantly under threat of infection. Recent studies revealed that some eukaryotic innate immune proteins are incredibly ancient, originating in the last common ancestor between bacteria and eukaryotes. Despite maintaining a few common features, the components and functions of immune pathways vary widely depending on the host organism. Why are these immune proteins so different, and how have they changed over evolutionary time? Toll/interleukin-1 receptor (TIR) signaling domains are ancient immunity modules that have been repeatedly repurposed across the tree of life. TIR domains from bacteria and plants induce immune signaling and/or direct death of infected cells by cleaving Nicotinamide Adenine Dinucleotide, a critical molecule for central metabolic processes. In contrast, animals primarily use TIR domains to initiate immune responses by acting as scaffolds, mediating protein-protein interactions to build immune signaling complexes. My work examines the function of TirA, a TIR-domain containing protein in Dictyostelium discoideum (Dd), which are phylogenetically located in between plants and animals. The way in which amoeba use TIR domains for immunity is poorly understood, partly because Dd lack most TIR-interacting proteins found in plant and animal immune systems. How are amoeba using TIR domains for their immunity? Are they using TIRs to perform novel functions or are they performing canonical biochemical functions with unique partner proteins? Understanding how amoeba use TIR domains for immunity will provide an important, phylogenetically-relevant puzzle piece to explaining how animal-type TIR signaling came to be and how eukaryotes innovate their immune defenses over evolutionary time. My preliminary results suggest that TirA functions enzymatically to induce immune responses. In future experiments, I will determine which TirA domains and residues mediate antibacterial responses such as bacterial predation and identify other Dd participants of the TirA pathway.

Title: Novel families of Chlamydiae bacteria form stable facultative endosymbiosis with wild Dictyostelium species

Authors: Mische Holland, Sarah Hesel, Tera Levin

Agenda

Affiliations: University of Pittsburgh, Dept of Biological Sciences

Endosymbiotic relationships between hosts and microbes have independently occurred all over the tree of life, ranging from antagonistic to mutualistic relationships. Phagocytic eukaryotes like amoebae constantly bring bacteria into their cells as they feed. While most of these bacteria get rapidly digested, amoebae also harbor a number of endosymbionts that escape host targeting. While sequencing an array of Dictyostelium amoebae isolates, we found that multiple strains harbored Chlamydia-like endosymbionts. First described by Haselkorn et al, 2021, these Dictyostelium-associated strains fall in unique clades within the Chlamydiae bacterial clade. We now present the first complete whole genomes of these Chlamydia-like endosymbionts. Preliminary analysis of these genomes show reduced genome size and lack of several key genes responsible for Chlamydiae biphasic lifestyle, suggesting that these symbionts are obligate intracellular symbionts. We found that antibiotic-curing the wild Dictyostelium strains of their endosymbionts did not affect amoeba growth, suggesting that these endosymbionts are stably inhierited, yet facultative. Next, we will be investigating if these strains can act as defensive symbionts during secondary Legionella infection. Through this work, we have produced high-quality whole genome assemblies of novel environmental Chlamydiae endosymbionts and will further investigate the impact of their residency in Dictyostelium amoeba hosts.

The pre-mRNA splicing kinase Prp4K modulates autophagy in *Dictyostelium*

Sabateeshan Mathavarajah[1], Elias B Habib[1], William D Kim[2], Megan M Aoki[2], Dale P Corkery[3], Kennedy IT Whelan[1], Jordan Lukacs[4], Jayme Salsman[1], Daniel Gaston[1], Robert J Huber[2]*, Graham Dellaire[1,5]*

Agenda

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Abstract

Pre-mRNA processing factor 4 kinase (PRP4K) is an essential protein in animal cells that regulates RNA splicing by facilitating the assembly of the spliceosome. Though essential, partial loss of *PRP4K* expression is associated with various forms of cancer suggesting that PRP4K may function as a tumour suppressor. The *Dictyostelium discoideum* ortholog of human PRP4K, Prp4k, is expressed throughout the life cycle, localizes to the nucleus and cytoplasm, and shares with PRP4K a highly conserved kinase domain that is required for splicing activity. To further study Prp4k function in *D. discoideum*, we used CRISPR/Cas9-mediated gene editing to generate the first *PRP4K* knockout model in any organism. Our characterization of *prp4k*$^-$ cells revealed several phenotypes during the life cycle, most notably, developmental defects associated with impaired autophagy. Autophagy defects were also observed when *PRP4K* was knocked down in human MCF-7 and *HeLa* cells. Mechanistically, loss of *PRP4K* reduced the expression and altered the splicing of the ESCRT-III gene *CHMP4B* in human cells and its ortholog *vps32* in *D. discoideum*. Re-introduction of *CHMP4B* or *vps32* cDNA in human cells and *D. discoideum*, respectively, suppressed the autophagy defects in *PRP4K*-deficient cells suggesting that the PRP4K-CHMP4B/Vps32 splicing circuit participates in the autophagic response and is conserved from *D. discoideum* to human. Based on these data, we then performed RNA sequencing of *prp4k*$^-$ cells to identify genes differentially expressed during growth and starvation (6h). GO term enrichment analyses revealed molecular and cellular processes impacted by the loss of *prp4k* and differentially expressed genes associated with phenotypes in *prp4k*$^-$ cells. Collectively, our findings establish *D. discoideum* as a valuable model system for studying PRP4K function and highlight genes and pathways of interest that can guide future work on this evolutionarily conserved splicing kinase.

Acute Stress and Multicellular Development Alter the Solubility of the *Dictyostelium* Sup35 Ortholog ERF3

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Among sequenced organisms, the genome of *Dictyostelium discoideum* is unique in that it encodes for a massive amount of repeat-rich sequences in the coding region of genes. This results in the *Dictyostelium* proteome encoding for thousands of repeat-rich proteins, with nearly 24% of the *Dictyostelium* proteome encoding Q/N-rich regions that are predicted to be prion-like in nature. To begin investigating the role of prion-like proteins in *Dictyostelium,* we decided to investigate ERF3, the *Dictyostelium* ortholog of the well-characterized yeast prion protein Sup35. Surprisingly, ERF3 lacks the Q/N-rich region required for prion formation in yeast, raising the question of whether this protein aggregates and has prion-like properties in *Dictyostelium*. Here, we found that ERF3 formed aggregates in response to acute cellular stress. However, unlike *bona fide* prions, these aggregates were not transmitted to progeny. We further found that aggregation of this protein is driven by the ordered C-terminal domain independently of the disordered N-terminal domain. Finally, we also observed aggregation of ERF3 under conditions that induce multicellular development, suggesting that this phenomenon may play a role in *Dictyostelium* development. Together, these findings suggest a role for regulated protein aggregation in *Dictyostelium* cells under stress and during development.

Using *Dictyostelium* to study the molecular impacts of disease-causing mutations in *CLN5*

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The neuronal ceroid lipofuscinoses (NCLs), collectively known as Batten disease, are neurological disorders that predominantly affect children but can arise at any age. Thirteen subtypes of Batten disease have been described; each one caused by a mutation in a genetically distinct ceroid lipofuscinosis neuronal (*CLN*) gene. All forms of Batten disease display the lysosomal accumulation of ceroid lipofuscin in cells of an affected individual. Homozygous mutations in *CLN5* cause a Batten disease subtype known as CLN5 disease. CLN5 is a soluble glycosylated protein that resides within the endoplasmic reticulum and lysosomes, as well as extracellularly. CLN5 participates in several cellular processes such as autophagy and lipid homeostasis, and previous work has speculated that CLN5 functions as a glycoside hydrolase, depalmitoylase, or bis(monoacylglycero)phosphate synthase. In this study, we used *Dictyostelium* as a model system to study the molecular impacts of disease-causing mutations in *CLN5*. By examining the alignment between human CLN5 and *Dictyostelium* Cln5, we identified five CLN5 disease-causing mutations that were present on conserved amino acids. We introduced those mutations into *Dictyostelium* Cln5 and generated cell lines expressing the mutated proteins in the *cln5*- genetic background. We observed that all mutations significantly reduce the secretion of Cln5, suggesting that Cln5 functions outside cells. In addition, we found that the disease-causing mutations affect lysosomal enzyme activity, as well as 20s proteasome activity. Finally, we show that disease-causing mutations differentially affect enzyme activity outside cells. Collectively, this study provides new insight into the function of CLN5 and the processes dysregulated by mutated CLN5 in humans.

Allorecognition, altruism and cheating in the *tgrB1-tgrC1-rapgapB* greenbeard pathway

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Agenda

A greenbeard genetic element encodes a rare perceptible signal, the ability to recognize the signal in others, and altruism towards other individuals that display the same signal. Greenbeards have been described in many organisms, but direct evidence for all the properties in one system has been scarce. The *tgrB1* and *tgrC1* genes of *Dictyostelium discoideum* encode two polymorphic transmembrane proteins which serve as an allorecognition system that protects cells from chimerism-associated perils. The TgrC1 protein functions as a ligand-signal and TgrB1 functions as its allotype-specific receptor. These genes have been described as a greenbeard system, but the evidence for altruism has been somewhat indirect. We have found that mutational activation of the *tgrB1* allele causes altruism. Mixing wild-type cells with cells that express an activated *tgrB1* allele increases wild-type spore production and relegates the mutant cells to the altruistic stalk. On the other hand, inactivation of *tgrB1* causes partial cheating. Mixing wild-type and *tgrB1*-null cells increases mutant spore production and wild-type stalk production. Moreover, the *tgrB1*-null cells cheat only on partners of the same *tgrC1*-allotype. Therefore, TgrB1 activation confers altruism whereas TgrB1 inactivation causes allotype-specific cheating. The greenbeard theory predicted the existence of falsebeards – hypothetical cheaters that display the greenbeard signal without being altruistic – but empirical evidence of falsebeards has been quite scarce. We have found that *rapgapB* is a component of the *tgrB1-tgrC1* signaling pathway because null-mutations in *rapgapB* function as mutual suppressors with null-mutations in *tgrB1* and *tgrC1*. We also found that inactivation of *rapgapB* causes partial cheating. The *rapgapB*$^{-}$ cells cheat on the wild type by avoiding the stalk fate and generating more spores than they do in pure populations. Moreover, *rapgapB*$^{-}$ cells cheat only on partners with compatible *tgrB1-tgrC1* allotypes, suggesting that greenbeard display and recognition are intact but decoupled from altruism. These findings provide a model system for the study of greenbeard signaling, altruism, and cheating in a genetically tractable system.

Title: TgrN1 mediates cell intercalation in D. discoideum

*Satoshi Kuwana1, Hidenori Hashimura1, Shoko Fujishiro1, Toyoko Sugita1, Tetsutaro Hayashi2, Mika Yoshimura2, Mariko Kuse2, Itoshi Nikaido2, Satoshi Sawai1

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During culmination of *D. discoideum*, cells undergo coordinated cellular movement and rearrangement to shape the final fruiting body anatomy. Starting from linearly ordered cell alignment along the antero-posterior axis, prestalk cells reposition themselves to form a two-layered tissue with radial symmetry. The underlying cell mechanics of this tissue patterning remain unresolved. Here, based on 3D light sheet imaging, we show that elongation of prestalk region as well as the stalk involves cell shape change, polarization and cell-cell intercalation. From full-length single-cell RNA-seq analysis, we identified a dozen or so paralogs of IPT-domain containing transmembrane adhesion protein Tgr whose patterns of expression and degradation correlated highly with the timing of the cell shape and polarity transitions. Based on observations of fluorescent protein tag, we show that these Tgr proteins are characterized into two types, ones that are distributed uniformly at the plasma membrane and the other that are membrane localized either along the anterior-posterior axis or inside-outside axis. Among the latter group, tgrN1 is unique in that it is only expressed in the stalk and enriched only at the stalk cell- stalk cell interface. TgrN1 knockout- strains were unable to intercalate properly in the stalk tube and thus failed to show tight cell packing. The resulting fruiting body had a stalk that was frequently bent in the middle despite stalk tube appearing normal with proper cellulose staining. TgrN1 knockout strain was defective both in the initial phase of cell intercalation that is accompanied by localized actin cytoskeleton and its regulators and the other without. The latter is characterized by inability of stalk vacuoles to mature and severely reduced cell swelling. In the knockout of transcription factor *STATa*, tgrN1 expression was not detectable, except for a very few founder cells in the PstAB core. On the other hand, TgrN1 was well expressed in a diguanylate cyclase (*dgcA*) null strain at the funnel-like and the stalk-like region below, however, these stalk cells were unable to polarize and intercalate. These data suggest that TgrN1 is a stalk-cell specific adhesion molecule essential for stalk elongation and that it works together with DgcA.

Allyson Sygro

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Agenda

PIP5K-Ras bistability initiates plasma membrane symmetry breaking to regulate cell polarity and migration

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Abstract
Symmetry breaking, polarity establishment, and spontaneous cell protrusion formation are fundamental but poorly explained cell behaviors. Here, we demonstrate that a biochemical network, where the mutually inhibitory localization of PIP5K and Ras activities plays a central role, governs these processes. First, in resting cells devoid of cytoskeletal activity, PIP5K is uniformly elevated on the plasma membrane, while Ras activity remains minimal. Symmetry is broken by spontaneous local displacements of PIP5K, coupled with simultaneous activations of Ras and downstream signaling events, including PI3K activation. Second, knockout of PIP5K dramatically increases both the incidence and size of Ras-PI3K activation patches, accompanied by branched F-actin assembly. This leads to enhanced cortical wave formation, increased protrusive activity, and a shift in migration mode. Third, high inducible overexpression of PIP5K virtually eliminates Ras-PI3K signaling, cytoskeletal activity, and cell migration, while acute recruitment of cytosolic PIP5K to the membrane induces contraction and blebs in cancer cells. These arrested phenotypes are reversed by reducing myosin II activity, indicating myosin's involvement in the PIP5K-Ras-centered regulatory network. Remarkably, low inducible overexpression of PIP5K unexpectedly facilitates polarity establishment, highlighting PIP5K as a highly sensitive master regulator of these processes. Simulations of a computational model combining an excitable system, cytoskeletal loops, and dynamic partitioning of PIP5K recreate the experimental observations. Taken together, our results reveal that a bistable, mutually exclusive localization of PIP5K and active Ras on the plasma membrane triggers the initial symmetry breaking. Coupled actomyosin reduction and increased actin polymerization lead to intermittently extended protrusions and, with feedback from the cytoskeleton, self-organizing, complementary gradients of PIP5K versus Ras steepen, raising the threshold of the networks at the rear and lowering it at the front to generate polarity for cell migration.

Investigating the functional relationship between Copine A (CpnA) and PatA, a Ca^{2+}-ATPase, in *Dictyostelium*

Cody T. Morrison[1], Amber D. Ide[1,2], Bridget K. Plude[1], Christer A. Carne[1,3], and Cynthia K. Damer[1]

Agenda

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Copines (Cpn) are a family of highly conserved, calcium-dependent membrane-binding proteins found in many eukaryotic organisms. *Dictyostelium discoideum* has six copine genes (*cpnA*-*cpnF*), and our lab has focused our studies on CpnA. CpnA binds to phosphatidylserine (PS) and translocates from the cytosol to the plasma membrane and intracellular organelles in response to increased intracellular calcium. *cpnA* knockout cells (*cpnA*-) exhibit many mutant phenotypes including defects in development, cytokinesis, membrane trafficking, chemotaxis, and adhesion. Using Fura-2 dextran and FITC-Lactadherin assays, we found that *cpnA*- cells have increased intracellular calcium concentrations compared to wild-type cells, which causes an increase in PS exposure. Chelating calcium with EGTA and BAPTA-AM decreased intracellular calcium concentrations and PS exposure in both *cpnA*- and wild-type cells. The increased intracellular calcium defect in *cpnA*- cells suggests a defect in a calcium pump. Previous research showed *cpnA*- cells made large contractile vacuoles (CVs) that could not be exocytosed properly compared to wild-type cells. We explored whether CpnA interacts with PatA, a Ca^{2+}-ATPase that localizes to the CV membrane. We found that CpnA co-precipitates with PatA. Western blotting revealed that the incubation of wild-type and *cpnA*- cells in media containing $CaCl_2$ resulted in increased PatA expression in both cell types, and the amount of PatA did not differ between wild-type and *cpnA*- cells. Using a *patA* antisense plasmid, we knocked down the expression of PatA in wild-type cells and found that the $patA^{KD}$ cells have similar phenotypes to *cpnA*- cells with increased intracellular calcium and PS exposure. $patA^{KD}$ cells also had a severe osmoregulation defect. When $patA^{KD}$ cells were placed in water, they made very large contractile vacuoles and became inviable over time. Overall, these results suggest that CpnA may interact directly with PatA on the surface of CVs to cause the activation of PatA.

Poster Presentations

Poster #1

<u>Title:</u> Identifying *L. pneumophila* genes required to circumvent *D. discoideum* nutritional immunity.

<u>Authors</u>: Danielle Farienella, Tera Levin
<u>Affiliations:</u> University of Pittsburgh, Dept of Biological Sciences

Amoeba predation can select for bacteria that are able to escape phagosome maturation and other innate immune mechanisms. Due to similarities between human and amoeba innate immunity, some bacteria that evolve to escape amoeba innate immunity can also evade human innate immunity. One example is *Legionella pneumophila*, the causative agent of Legionnaires' disease. This project investigates the genes in *L. pneumophila* required for infection of *D. discoideum*, with a specific focus on how *L. pneumophila* overcomes iron starvation within host cells. In both *D. discoideum* and humans, one important mediator of pathogen iron starvation is the iron transporter *Nramp*. We are using parallel Tn-seq screens to identify a comprehensive list of genes *L. pneumophila* uses to infect *D. discoideum* and determine which specific genes may have genetic interactions with *Nramp*. This information may help to reveal the similarities and differences in nutritional immunity between amoebae and macrophages. It will also help us to understand how amoeba selective pressures may have contributed to the pathogenesis of *L. pneumophila* in humans.

Poster #2

Single objective light sheet fluorescence microscopy greatly reduces phototoxicity and photobleaching and increases time resolution when imaging live cells.

Rachel Stadler[1], Maria Traver[2], Tanner Fadero[3], and Joseph Brzostowski[2]

[1] NIH, NIAID, Laboratory of Malaria and Vector Research
[2] NIH, NIAID, Laboratory of Immunogenetics, Twinbrook Imaging Facility
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We are building a single objective light sheet (SOLS) microscope to image individual live cells expressing fluorescently tagged proteins. The design of our system incorporates the specialized “Snouty” objective lens to capture the remote image formed by the system’s second objective and nearly the full numerical aperture of the primary objective. The Snouty/SOLS microscope provides near confocal resolution, acquires 100 slice/30 micron volumes in 100 milliseconds at its fastest frame rate and employs a standard inverted microscope platform, making it available for use with common cell imaging chambers and incubation systems. Details of the design, comparisons to other imaging modalities, and what it’s like for a “biologist-by-training” to construct a state-of-the-art, home-built instrument will be discussed.

Poster #3

Dioxygenases that regulate *Dictyostelium* as potential oxygen sensors

Mitchell Rich[1], Hanke van der Wel[1], Julie Hesnard[2], Christophe Anjard[2], and Christopher M. West[1]

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Growing attention is being given to post-translational modifications by dioxygenases as oxygen sensors. The discovery in animals that prolyl hydroxylase domain protein 2 (PHD2) destabilizes the co-transcriptional factor HIF1α provided the first molecular mechanism for oxygen sensing, and was followed by finding that its likely ortholog in protists (PhyA) regulates E3(SCF)Ub-ligases via modification of its Skp1 subunit. These findings raise the question of the roles in physiological oxygen sensing of the dozens of other dioxygenases that modify protein amino acids or demethylate them. The social amoeba *Dictyostelium* also exhibits oxygen-dependent behavior at the single cell (aerotaxis) and multicellular levels, and the finding that PhyA controls the oxygen checkpoint for culmination but whose absence can be overridden by hyperoxia inspired a search for the contributions of other dioxygenases in a simplified model where their interactions might be teased out. We employed CRISPR-Cas9 to knock out 16 of its 19 predicted protein dioxygenases, and submitted them to our culmination and aerotaxis assays. As for PhyA, knockouts of ADO, PHA, and JcdD increased the oxygen threshold for culmination to varying extents. ADO (2-aminoethanethiol dioxygenase) is known to mediate oxygen-dependent protein degradation of proteins with a sub-N-terminal Cys-residue via an N-end rule pathway in plants and animals. Knockout of PHA (predicted prolyl 4-hydroxylase) also renders aberrant stalk cell differentiation. JcdD is a homolog of the human arginine hydroxylase JmjD5. In addition, knockout of the Jumonji-C containing protein JcdG resulted in a slow growth phenotype that was rescued by secondary CRISPR-Cas9, but did not affect culmination. In contrast, none of the 16 mutants appeared to affect single cell aerotaxis. Future studies will confirm the specificity of the three knockouts and probe the role of their dioxygenase activities *per se*, which will set the stage for investigating their biochemical activities and whether they act in concert or independently for different cellular functions and cell types.

Poster #4

Novel Antisera Reveal New Insights into the Modification of Nucleocytosolic Proteins in *Dictyostelium* and Other Protists with O-Fucose

Megna Tiwari[1,2,3], Jen Teal[1], Elisabet Gas-Pascual[1,2,3], Msano Mandalasi[1,2,3] Manish Goyal[4], Marla Popov[5], Kenjiroo Matsumoto[2,8], Marianne Grafe[6], Ralf Graf[6], Robert S. Haltiwanger[2,3], Neil Olszewski[7], Ron Orlando[2,3,5], John Samuelson[4], and Christopher M. West[1,2,3]

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Cellular adaptations to change often involve post-translational modifications of nucleocytoplasmic proteins. Of interest here is the O-fucosyltransferase (OFT) SPY, named after the *Arabidopsis* OFT SPINDLY, that modifies Ser/Thr residues with a single fucose (O-Fuc) in the social amoeba *Dictyostelium*, the pathogen *Toxoplasma gondii* and higher plants. Furthermore, OFT-like genes exist in many other protists and pathogens such as *Acanthamoeba* and *Cryptosporidium*. O-fucosylation is required for optimal proliferation of *Dictyostelium and Toxoplasma* and is paralogous to the O-GlcNAcylation of nucleocytoplasmic proteins of plants and animals, which is involved in stress and nutritional responses. O-Fuc was first discovered in *Toxoplasma* using *Aleuria aurantia* lectin (AAL) but its broad specificity for any terminal fucose residue on N- and O-linked glycans generated in the secretory pathway limits its use. To facilitate investigation of O-Fuc in fucose-rich organisms like *Dictyostelium*, we developed antibodies specific for fucose-O-Ser and fucose-O-Thr (anti-FOS/T). Unlike AAL, anti-FOS/T only detects nucleocytoplasmic proteins modified by OFT. Anti-FOS and anti-FOT label *Dictyostelium, Toxoplasma*, and *Acanthamoeba* nuclei in a pattern reminiscent of O-GlcNAc in animal cells. Western blotting confirms its utility in *Arabidopsis* and shows that the *Dictyostelium* O-fucome is highly induced during starvation-induced development to form slugs and fruiting bodies. Analysis of anti-FOS and anti-FOT pull-downs from *Dictyostelium* in a mass spectrometry-based proteomics workflow yielded an expanded O-fucome compared to AAL analysis of a cytosolic fraction. Relative to OFT-KO cells, anti-FOS and anti-FOT captured partially overlapping sets of proteins that correlated with the presence of Ser-rich and Thr-rich sequences. Separate partially overlapping sets of proteins were captured from developing slugs, suggesting developmental regulation. The *Dictyostelium* O-fucome partially overlaps with the O-fucomes of *Toxoplasma* and *Arabidopsis,* including proteins associated with DNA replication, transcription, translation, the ubiquitin proteasome system, nuclear transport, metabolism and the nuclear pore complex. By epitope-tagging a *Toxoplasma* GPN-GTPase gene, we confirmed a high level of O-fucosylation in its Ser-rich domain, and that O-fucosylation is required to achieve normal expression levels. We speculate that protist O-Fuc is responsive to external signals such as nutrition, which may control the availability of GDP-Fuc required by OFT to modulate the stability of key proteins with disordered Ser- or Thr-rich domains.

Poster #5

Phenotypic Characterization of Copine D Mutants in *Dictyostelium*

Bridget Plude, Cody Morrison, Sonya Ruiz, Sela Damer-Daigle, Jordyn Anklam, and Cynthia Damer
Central Michigan University, Mount Pleasant, Michigan USA

Genetic screenings and functional studies indicate that copine proteins are highly expressed in human cancer cells and promote cellular proliferation, migration, and metastasis. Copines are a family of highly conserved, calcium-dependent membrane-binding proteins found in most eukaryotes. The human genome contains nine copine genes that are differentially expressed in various tissue types. However, the cellular function of copines is not well understood, and the cellular mechanisms in which copines are involved in human disease and cancer progression are unknown. We are using *Dictyostelium discoideum*, which has six copines genes, as a model organism to investigate the function of copines. We obtained two insertional *cpnD* mutants and the parental AX4 cell line from the *Dictyostelium* REMI mutant project to investigate the function of *cpnD*. The two *cpnD* mutant strains have the REMI insertions in different sites within the *cpnD* gene: at 459 bp in *cpnD-50* and at 291 bp in *cpnD-168*. We made clonal populations of each strain and then verified the site of the REMI insertion within the *cpnD* gene using PCR. We then characterized the phenotypes of the two *cpnD* mutants focusing on phenotypic defects we previously observed in *cpnA* and *cpnC* null mutants. In general, the *cpnD-168* mutant strain exhibited more severe phenotypes than the *cpnD-50* mutant. Similar to *cpnC-* cells, *cpnD* mutants were less adhesive and had less SibA expression than parental AX4 cells. While *cpnA-* and *cpnC-* cells exhibited normal growth rates, *cpnD* mutants grew faster in axenic culture than the parental AX4 cells. *cpnD* mutants appeared to develop with normal timing and morphology, while *cpnA-* and *cpnC-* cells exhibited distinct developmental phenotypes. Unlike *cpnA-* cells, the mutants showed no significant difference in contractile vacuole size compared to AX4 cells when placed in water. However, like *cpnA-* cells, *cpnD* mutants had increased PS exposure.
Our data shows that *cpnD* mutants have distinct phenotypes from *cpnA* and *cpnC* mutant cells, and indicates that all three copines have distinct functions. Our objective is to determine the biological functions of each copine gene. This knowledge is potentially valuable in understanding the function of the evolutionarily conserved copine genes and their roles in human cancers.

Poster #6

Back to Basics: Application of the Halo and Well Assays to Analysis of Negative Chemotaxis in *Dictyostelium*

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The widespread use of *Dictyostelium* as a model system for chemotaxis frequently relies on some combination of pumps to dispense pulses of chemoattractant, customized chemotactic chambers, and/or imaging systems that allow visualization and tracking of individual cells, equipment that is not universally available, particularly at primarily undergraduate institutions or in laboratories that do not specialize in the study of chemotaxis. Here we report the successful application of two "classical" chemotaxis assays (the halo and well assay) to analysis of potential instances of negative chemotaxis. First, a halo assay, in which drops of starved cells were plated on agar containing 10 µM cAMP and various amounts of the reported chemorepellant 8-CPT-cAMP, was used to assess the effect of 8-CPT-cAMP on the outward migration of wild-type cells (*Ax2*, *Ax3*) and mutants with a known decrease (*gskA*$^-$, *statA*$^-$) or increase (*cadA*$^-$) in chemotactic ability. We were able to demonstrate dose-dependent inhibition of outward migration when 8-CPT-cAMP was present at 50- or 100-fold excess relative to cAMP but did not observe increased or decreased sensitivity to inhibition by 8-CPT-cAMP in any mutant strain relative to its wild-type parent. Differences between strains in the fraction of cells responding and/or density of cells at various distances outside the initial drop, independent of 8-CPT-cAMP levels, were noted but not quantified. Hence, the halo assay may also have promise for detection of some abnormalities in positive chemotaxis. In a separate study, we sought to extend the study of negative chemotaxis to signaling dependent on folate and the fAR1 receptor by employing drops of unstarved *Ax2* cells spotted on agar containing a constant concentration of one potential receptor agonist, in addition to a central well in which another receptor agonist was loaded (in order to create a gradient of the latter via diffusion). Both the relative area of migration (area of final spot/area of initial spot) and net migration (outward movement toward the well minus any movement away from the well) were recorded. Under conditions in which 100 µM folate, the normal activator of the receptor, was introduced at a constant level and a gradient of a weaker chemoattractant (methotrexate, L-methyl folate, or pterine-6-COOH) was present, net negative migration was observed; gradients of methotrexate or pterine-6-COOH also appeared to inhibit the relative area of migration. While further optimization of relative chemoattractant concentrations would be necessary to conclusively demonstrate negative chemotaxis, these outcomes are consistent with a recently advanced

model for the ability of two receptor agonists to produce apparent chemorepulsion via competition for a common receptor. Overall, we conclude that, in spite of their limitations, these less equipment-intensive assays remain valuable, both as educational tools and in the generation of novel data.

Poster #7

An Alphafold3 based structural modeling to predict GPCR-G protein coupling

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Abstract GPCRs couple to specific G-proteins to achieve appropriate functional responses. Here, we are using Alphafold3 to model structures of GPCR, Gα, and Gβγ _that predict possible coupling between a GPCR and various G protein subunits. We use the cyclic AMP receptor (cAR1) as a model to predict its potential coupling with various Gα _and Gβγ _subunits. It has been believed that cAR1 couples with Gα2Gβγ _and but not Gα4Gβγ _to mediate signaling. Interestingly, our models of cAR1 with Gα2Gβγ _or Gα4Gβγ _suggest that cAR1 can couple with both Gα2Gβγ _and Gα4Gβγ. We will experimentally test whether cAR1 can couple with Gα4Gβγ _to mediate cAMP-triggered signaling, and thus test the validity of the method.

Poster #8

How prototypical phagocytes detect and respond to various bacteria?

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The social amoebae *Dictyostelium discoideum* are prototypical phagocytes that chase various bacteria via chemotaxis and consume them as food via phagocytosis. Previously, we discovered that *D. discoideum* amoebae use a C-class GPCR, fAR1, to detect folic acid released from bacteria for chemotaxis to catch various bacteria. Upon catching them, the amoeba uses fAR1 to recognize LPS on the surface of gram-negative bacteria. This finding suggests that a GPCR-mediated signaling network controls reorganization of the actin cytoskeleton for both chemotaxis and phagocytosis. C-class GPCRs have a common structure consisting of an N-terminal extracellular Venus flytrap (VFT) domain for ligand recognition and binding. The genome encodes 14 C-class GPCRs. The VFT domain on one of them, fAR1, detects a chemoattractant, folic acid, and the sugar part of LPS, which is a major microbe-associated molecular pattern (MAMP). Recently, we discovered that LTA, a molecule on the surface of gram-positive bacteria, could induce ERK phosphorylation in D. discoideum, and this LTA-induced ERK-P requires Gα4 and Gβγ subunits. In addition, LTA failed to induce ERK-P in GrlL (fAR1) null and GrlG null, indicating that both GrlL and GrlG are required for LTA-mediated signaling. Using AlphaFold 3-based protein structure modeling, we found that GrlL and GrlG can form a heterodimer that is associated with Gα4 and Gβγ, which detect lipoteichoic acids (LTA) on the surface of gram-positive bacteria. Thus, our findings indicate that C-class GPCRs represent a new class of pattern recognition receptors (PRR) utilized by prototypical phagocytes to detect and eat various bacteria.

Poster #9

Title: ***Dictyostelium discoideum* as a chassis organism for developing eukaryotic biosensors**

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Dictyostelium discoideum is a free-living amoeba found in soil and leaf litter capable of surviving in many environments, including distilled water. As a model system in biology, *D. discoideum* is genetically tractable and has a large foundation of techniques and resources for research (i.e. dictybase.org, GoldenBraid modular cloning toolkit, etc.). These traits make *D. discoideum* an ideal chassis organism for developing synthetic biology sensors in a eukaryotic system. We aim to use *D. discoideum* to develop eukaryotic biosensors using membrane-bound proteins including G-protein coupled receptors (GPCRs) and insect olfactory receptors. GPCR pathways, such as the cyclic AMP (cAMP) receptors, have been heavily studied *D. discoideum* for cell-to-cell communication for motility. Using CRISPR Cas9 we have begun knocking out genes in native GPCR pathways. We are currently cloning and expressing engineered human GPCRs with their cognate G-proteins to test for functionality in *D. discoideum*. Another source of potential biosensors are insect olfactory receptors, which are ligand gated ion channels. We have cloned and expressed insect olfactory receptors in *D. discoideum* using the Dicty GoldenBraid toolkit. To test activation of insect olfactory receptors we grew *D. discoideum* on Indium tin oxide (ITO) coated glass slides and measured changes in electrical conductance after the addition of the receptor's ligand.

Poster #10

Structural and Regulatory Features of the D. discoideum Cyclic Dinucleotide Cyclase DgcA

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In prokaryotes, cyclic dinucleotides (cDNs) are second messenger molecules that participate in a great diversity of signaling pathways. Among the most conserved bacterial dinucleotide cyclases is the GGDEF (Gly-Gly-Asp-Glu-Phe) family of enzymes, which synthesize 3’3’-cyclic-diguanosine-monophosphate (c-di-GMP). cDNs were believed to be exclusively prokaryotic until the first eukaryote GGDEF-family cyclase, DgcA, from the social amoeba Dictyostelium discoideum, was identified; this discovery paved for the identification of the DncV-family of cyclic- GMP-AMP synthases in animal cells, which play a protective role in the innate immune response to double stranded nucleic acids (dsDNA or dsRNA). In D. discoideum, synthesis of c-di-GMP by DgcA is essential for stalk differentiation during late development, regulating the stalk-specific Protein Kinase A (PKA)-dependent transcriptional program in synergy with the hexanophenone morphogen, Differentiation Inducing Factor 1 (DIF-1). While the mRNA expression of DgcA peaks after 10h of starvation, high levels of c-di-GMP are detected after more than 20h of starvation in the tip, where stalk formation initiates, suggesting that DgcA enzymatic activity may increase under specific conditions. To gain insight into the molecular mechanisms involved in DgcA regulation, we first generated a structural model of the DgcA protein. We found that, in addition to the previously identified GGDEF enzymatic domain, DgcA contains a hydrophobic putative transmembrane helix and an N-terminal Per-Arnt-Sim (PAS) domain, which has been reported to act as a sensor or interaction module in some bacterial GGDEF cyclases. Interestingly, nearly half of the DgcA protein is predicted to be intrinsically disordered, suggesting that it may adopt a specific structure only upon binding a molecule or other proteins. Additionally, AlphaFold modelling indicated that DgcA may form a dimer, a configuration that could be relevant for its catalytic activity. In agreement with our model, overexpressed DgcA localized to intracellular membranes and is enriched in membrane fractions of growing D. discoideum. Furthermore, we found that the C-terminal GGDEF-containing domain of DgcA mediates its constitutive dimerization, and the identified N-terminal region, containing the PAS domain is essential for its enzymatic activity. Moreover, we found that c-di-GMP concentration in growing D. discoideum was regulated by nitrogen availability and by hyperosmotic stress, but not by dsDNA or dsRNA. Our data suggest that DgcA may act as a transmembrane sensor protein regulated by changes in nutrient availability and in cellular turgor, shedding light into the complexity of cyclic dinucleotide signaling in eukaryotes.

Poster #11

Mutual suppression between mutations in the *Dictyostelium* greenbeard pathway restores wild-type development

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The greenbeard allorecognition pathway of *Dictyostelium discoideum* is mediated by the polymorphic transmembrane proteins TgrB1 and TgrC1. TgrC1 is a ligand and TgrB1 is its receptor, and together they mediate allorecognition with accompanying altruism during the transition from unicellular to multicellular development. A genetic screen for suppressors of developmental defects in the non-compatible allotype *tgrB1-C1* pathway revealed activating mutations in *tgrB1* and inactivating mutations in *rapgapB*, which encodes a regulator of the small GTPase protein RapA. Inactivation of either *tgrB1*, *tgrC1*, or *rapgapB* leads to developmental defects. However, surprisingly, the respective double-mutant strains *tgrB1⁻rapgapB⁻* and *tgrC1⁻rapgapB⁻* develop well and produce wild-type-like fruiting bodies. This mutual suppression could result from the induction of an alternative developmental pathway or restoration of the wild-type pathway, but morphological analyses alone could not resolve this question. Here we show that the mutual suppressors restore wild-type development. We used RNA-sequencing analyses to compare the transcriptomes of the wild type to those of six mutant strains and found that the single-gene mutations resulted in attenuated transcriptome progression over developmental time, whereas the double-gene mutation strains exhibited near wild-type transcriptomes. We also analyzed a strain that carries an activated *tgrB1* allele in the wild type and found evidence for interactions between the activated and the resident alleles. Our findings suggest that *tgrB1*, *tgrC1*, and *rapgapB* are involved in a regulatory feedback pathway in which *rapgapB* negatively regulates *tgrB1* and *tgrC1* whereas *tgrB1* and *tgrC1* positively regulate *rapgapB*. These findings suggest that the *D. discoideum* greenbeard pathway interfaces with the central RapGAPB-RapA regulatory pathway, providing molecular insight into a mutual suppression mechanism in which combining two deleterious mutations leads to the restoration of wild-type behavior.

Poster #12

Investigating the functional relationship between Copine A (CpnA) and PatA, a Ca^{2+}-ATPase, in *Dictyostelium*

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Copines (Cpn) are a family of highly conserved, calcium-dependent membrane-binding proteins found in many eukaryotic organisms. *Dictyostelium discoideum* has six copine genes (*cpnA*-*cpnF*), and our lab has focused our studies on CpnA. CpnA binds to phosphatidylserine (PS) and translocates from the cytosol to the plasma membrane and intracellular organelles in response to increased intracellular calcium. *cpnA* knockout cells (*cpnA*-) exhibit many mutant phenotypes including defects in development, cytokinesis, membrane trafficking, chemotaxis, and adhesion. Using Fura-2 dextran and FITC-Lactadherin assays, we found that *cpnA*- cells have increased intracellular calcium concentrations compared to wild-type cells, which causes an increase in PS exposure. Chelating calcium with EGTA and BAPTA-AM decreased intracellular calcium concentrations and PS exposure in both *cpnA*- and wild-type cells. The increased intracellular calcium defect in *cpnA*- cells suggests a defect in a calcium pump. Previous research showed *cpnA*- cells made large contractile vacuoles (CVs) that could not be exocytosed properly compared to wild-type cells. We explored whether CpnA interacts with PatA, a Ca^{2+}-ATPase that localizes to the CV membrane. We found that CpnA co-precipitates with PatA. Western blotting revealed that the incubation of wild-type and *cpnA*- cells in media containing $CaCl_2$ resulted in increased PatA expression in both cell types, and the amount of PatA did not differ between wild-type and *cpnA*- cells. Using a *patA* antisense plasmid, we knocked down the expression of PatA in wild-type cells and found that the *patA*^{KD} cells have similar phenotypes to *cpnA*- cells with increased intracellular calcium and PS exposure. *patA*^{KD} cells also had a severe osmoregulation defect. When *patA*^{KD} cells were placed in water, they made very large contractile vacuoles and became inviable over time. Overall, these results suggest that CpnA may interact directly with PatA on the surface of CVs to cause the activation of PatA.

Poster #13

Impact of GATA Transcription Factors on *Dictyostelium* Cell Fates

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Abstract

Mitogen-activated protein kinases (MAPKs) mediate many different cellular responses, yet atypical MAPKs, such as Erk2 in *Dictyostelium*, remain understudied. Our previous work demonstrated that Erk2 is required for chemotaxis and regulates the translocation of GtaC, a GATA transcription factor (TF) that plays a key role in multicellular development. In response to chemoattractant stimulation, the phosphorylation of GtaC by Erk2 results in a rapid translocation of GtaC into the cytoplasm. Erk2 is essential for multiple chemotactic responses (folate and cAMP stimulation) that regulate distinct cell fates such as foraging and developmental aggregation. The involvement of Erk2 regulation of GtaC in different cell fates suggests that other TFs might contribute to cell-fate specific gene expression. To investigate this possibility, we have extended our analysis to explore the regulation and function of three additional GATA TFs (GtaN, GtaO, and GtaS) which also possess Erk2-preferred phosphorylation sites akin to those found in GtaC. Our initial analysis of GFP fusions to these TFs indicated that all of them are localized to the nucleus. Expression of the GFP-GtaN fusion resulted in a significant delay in *Dictyostelium* development, suggesting a possible role in the regulation of early developmental processes. We are currently examining the function of the zinc-finger (DNA binding domain) and putative Erk2 phosphorylation sites in the GFP constructs of these three TFs through mutational analysis. We are also in the process of creating gene disruption mutants for these TFs. Moreover, we are investigating the potential regulation of these TFs in response to chemotactic stimulation. These findings will help us to understand the role of Erk2-regulated TFs in cell fate.

*Poster presenter

Poster #14

High-resolution measurement of intracellular signaling mechanisms using giant cells that retain cell-to-cell signaling

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Disturbances in signal transduction are the cause of many diseases, including cancer, atherosclerosis, and immune disorders. The details of the signaling mechanism are needed to be elucidated.
Dictyostelium discoideum is a model organism for signal transduction. However, the cell size of *D. discoideum* is too small to observe the details of intracellular signal transduction. Limitations in temporal and spatial resolution due to the cell size limitations have hampered the elucidation of intracellular signaling mechanisms.
To increase the resolutions, we established a method for constructing giant cells that maintain cell-to-cell signaling by inhibiting cytokinesis of *D. discoideum* cells [1]. We used confocal fluorescence microscopy and highly sensitive fluorescent probes to observe cAMP signaling and the associated changes in Ca^{2+} signaling. The results showed that the signals are synthesized and secreted by propagating from front to rear in a single chemotactic cell. Fluorescent labeling of the front and rear sides of the cells confirmed that the front and rear polarity was maintained even in giant cells. The use of giant cells enables the observation of intracellular signal transduction with a spatio-temporal resolution 10 times higher than normal.

[1] Hayashida *et al.*, (2024) Biophysical Journal. doi: 10.1016/j.bpj.2024.07.017.

Poster #15

Pattern formation along signaling gradients driven by active droplet behaviour of cell groups

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Abstract:
Gradients of extracellular signals organise cells in tissues. Although there are several models for how gradients can pattern cell behaviour, it is not clear how cells react to gradients when the population is undergoing 3D morphogenesis, in which cell-cell and cell-signal interactions are continually changing. *Dictyostelium* cells follow gradients of their nutritional source to feed and maintain their undifferentiated state. Using light sheet imaging to simultaneously monitor signaling, single cell and population dynamics, we show that the cells migrate towards nutritional gradients in swarms. As swarms advance, they deposit clumps of cells at the rear, triggering differentiation. Clump deposition is explained by a physical model in which cell swarms behave as active droplets: cells proliferate within the swarm, with clump shedding occurring at a critical population size, at which cells at the rear no longer perceive the gradient and are not retained by the emergent surface tension of the swarm. The droplet model predicts vortex motion of the cells within the swarm emerging from the local transfer of propulsion forces, a prediction validated by 3D tracking of single cells. This active fluid behaviour reveals a developmental mechanism we term “musical chairs” decision-making, in which the decision to proliferate or differentiate is determined by the position of a cell within the group as it bifurcates.

Poster #16

The mechanism by which dictyostelids repel plant-parasitic nematodes

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We are interested in the chemical communications between *Dictyostelium* and other soil microbes. We found that *Dictyostelium* can repel plant-parasitic nematodes. Plant-parasitic nematodes parasite a wide range of plants, including several major crop plants, and it causes huge agricultural losses. Our previous study showed that the cell extracts of *D. discoideum* can repel one of the plant parasitic nematodes *Meloidogyne incognita* and protect plant root from its infection. (Saito YF et al., PLoS ONE 12(9):e0204671, 2018)

As we have reported before, we found that the conditioned medium (CM) of *D. discoideum* can repel *M. incognita*. To identify the repellent substances from *D. discoideum*, CM was subjected to metabolomics analysis and 126 chemical compounds were identified. We analyzed them by *in vitro* repellent assay and found that 13 of them had repellent activity towards *M. incognita*. Four of these are amino acids, and the others have antioxidant properties. Especially, basic amino acids had strong repellent activity towards *M. incognita*.

To confirm *M. incognita* recognizes these substances from *D. discoideum* , we tried to identify receptors in the phasmid of *M. incognita* . The chemosensory system of *M. incognita* has not been extensively investigated yet . We knocked down 54 genes including GPCR and performed the repellent assay. We found several GPCRs are related to detect repellent substances from *D. discoideum*.

This study demonstrated that 13 substances including basic amino acids derived from *D. discoideum* are keys to repelling *M. incognita*, and confirmed *M. incognita* recognizes these substances with several GPCRs.

Poster #17

Studies of Biological Noise in Stress Response in *Dictyostelium discoideum*

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Biological noise refers to nongenetic cell-to-cell variation that arises from identifiable differences between cells (i.e., extrinsic biological noise) or stochastic variation in biochemical reactions (i.e., intrinsic biological noise). We are interested in identifying the major sources of biological noise in stress responses. Using microfluidic devices, we conduct single-cell analysis of individual cell contents to measure variation in oxidative stress between individual *D. discoideum* cells, then perform statistical evaluation of the single-cell measurements. Most recently, we have demonstrated that cell cycle position is a major contributor to cell-to-cell variation in oxidative stress, and our results suggest that this effect may be mediated by mitochondrial function. In tandem, we are developing methods to examine heterogeneity in cell signaling through the PI3K-PKB pathway during social development. The methods used in these experiments are a useful complement to ensemble measurements by Western blotting and to traditional single-cell measurements by microscopy or flow cytometry.

Poster #18

Identifying the binding partners of FszA and FszB in *Dictyostelium discoideum*

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Mitochondrial dynamics (fusion, fission, and motility) are important to the homeostasis of the cell and play a role in mitochondrial related diseases such as Alpers disease, and neurodegenerative diseases such as Parkinson's disease. However, the direct connection between mitochondrial dynamics and neurodegenerative diseases remains unclear. *Dictyostelium discoideum* is a mitochondrial disease model system, and our lab has established it as a model for studying mitochondrial dynamics.

This study investigates the binding partners of two proteins, FszA and FszB, which are hypothesized to be involved in *D. discoideum* mitochondrial dynamics. To carry out this project, we must first establish a mitochondrial isolation protocol. Using FLAG-FszA cells, several lysis methods were tested followed by various differential centrifugation protocols. Using porin as a marker, western blotting was carried out to confirm enrichment of mitochondria. Preliminary results indicate that filter lysis is more effective than homogenization with a Potter-Elvehjem homogenizer. A simple 3 step differential centrifugation process using 600xg and 10,000xg results in mitochondrial enrichment in the appropriate fraction.

Finally, we will create a FLAG-FszB expressing strain and carry out co-immunoprecipitations using anti-FLAG magnetic beads to harvest FszA/B proteins along with their binding partners which will be identified by mass spectrometry. Identified proteins are expected to have roles along with FszA/B in fission, fusion and motility of the mitochondria. Identification of FszA/B binding partners will contribute to our understanding of mechanism used by *D. discoideum* to carry out mitochondrial dynamics and eventually into diseases effected by defects in mitochondrial dynamics.

Poster #19

Stress-Specific Protein Aggregation in *Dictyostelium discoideum*: Unraveling PolyQ Dynamics and the Role of PPK1

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Protein aggregation is a double-edged sword in cellular biology, serving as both a protective response to stress and a potential pathway to dysfunction. In *Dictyostelium discoideum*, we investigated how environmental stressors—specifically heat and multicellular development—shape protein aggregation and solubility. Using mass spectrometry, we identified 1,116 proteins that became insoluble during heat stress and 456 proteins that aggregated during developmental stages. These patterns were distinctly stress-specific; heat-stressed cells primarily aggregated proteins involved in DNA repair, ribosomal processing, and kinase activity, while developmental conditions favored the aggregation of calcium-binding and cell surface proteins.

Our study also focused on polyglutamine (polyQ) proteins, which are often linked to neurodegenerative diseases like Huntington's. We found that polyQ proteins exhibited significant aggregation under nutrient stress and during multicellular development, indicating their heightened sensitivity to environmental conditions. Furthermore, we identified polyphosphate kinase 1 (PPK1) as a critical modulator of aggregation dynamics. Knocking out PPK1 resulted in significant shifts in aggregation patterns, particularly among proteins with prion-like domains, suggesting that PPK1 plays a vital role in maintaining the balance between functional and potentially toxic aggregates.

Overall, our findings reveal that protein aggregation in *Dictyostelium* is a finely tuned, stress-specific process essential for cellular adaptation. The interactions between polyQ proteins and PPK1 underscore their protective functions and highlight the delicate balance between stability and toxicity in cellular environments.